

Tri-plane diagrams for simple surface knots

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- 2 Surface Knots
- 3 Classifying Unknots
- 4 Compiling a List of Known Knotted Surfaces
- 5 Conclusion

Introduction

Knot Theory

Definition

Knot theory: A subsection of topology where mathematical knots and their properties are studied

Some common knots

The following are common knots typically observed in 3 dimensions



Figure: From left to right, an unknot, a trefoil, and a figure 8

Planar isotopy

Definition

Planar isotopy is performed to change the projection of a knot without changing any of its qualities

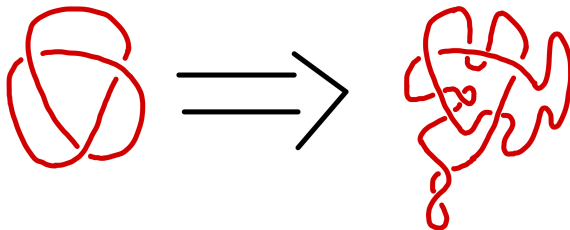


Figure: 2 projections of a trefoil knot

Reidemeister moves

Definition

Planar isotopy is performed using Reidemeister moves

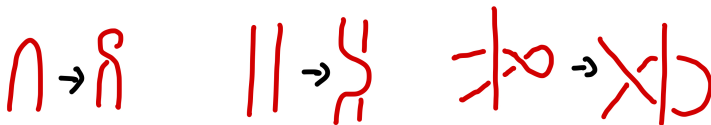


Figure: R_1 , R_2 , and R_3 moves

Links

Definition

A link is a set of 2 or more knots linked together



Figure: 2 trefoil knots linked

Knots in higher dimensions

Theorem

All knots are equivalent to the Unknot in 4 dimensions or greater.

Surface Knots

Knots in Four-Space

There isn't much to say about one-dimensional knots in $\mathbb{R}^4$, but we are interested in studying two-dimensional knots, which we will call **surface knots** (or **knotted surfaces**), in four-space. We can depict these surface knots with three planar diagrams, which we call **tri-plane diagrams**.

Tri-plane Diagrams

Definition

A **tri-plane diagram** $K = (K_1, K_2, K_3)$ must satisfy the following.

- Each diagram should have the same number of strands (and hence same number of endpoints on the horizontal axis).
- Each strand should be a trivial tangle, that is, each strand has one relative maximum w.r.t the horizontal axis.
- Each of $K_1 \cup \overline{K_2}$, $K_2 \cup \overline{K_3}$, and $K_3 \cup \overline{K_1}$ should be diagrams for an unlink (where $\overline{K_i}$ is the mirror over the horizontal axis).



Figure: 1. An example of a tri-plane diagram $K = (K_1, K_2, K_3)$.

Tri-plane Diagrams (continued)

One can verify that the diagram in Figure 1 is a tri-plane diagram. In fact, it depicts the knotted surface known as the spun trefoil.

Clearly, each tangle is trivial and each diagram has four strands. Inspecting the below figure allows us to verify that each mirrored union is an unlink.

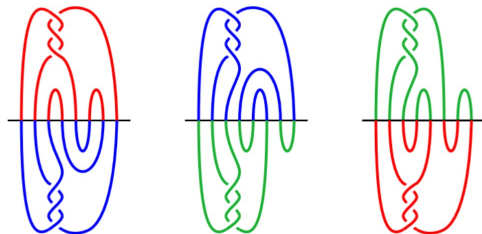


Figure: 2. A verification that the mirrored unions are unlinks.

Tri-plane Moves

Just as in 3-dimensional knot theory, tri-plane diagrams can undergo local changes via the three Reidemeister moves, which are pictured in Figure 3.

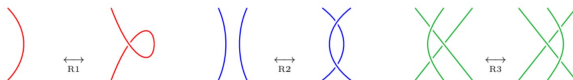


Figure: 3. The three Reidemeister moves.

Additionally, we can perform mutual braid moves, which are shown in Figure 4.



Figure: 4. A mutual Braid Move

Tri-plane Moves (continued)

We can also perform stabilization and destabilization moves as pictured in Figure 5.

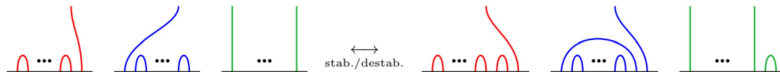


Figure: 5. Stabilization and Destabilization

Definition

A **tri-plane move** is a manipulation of a tri-plane diagram via a Reidemeister move, a mutual braid move, or a stabilization/destabilization.

Knotted Surfaces

Definition

We write $K \sim K'$ if K' can be obtained from K via a finite series of tri-plane moves and planar isotopies. One can check that $\sim$ as defined above is an equivalence relation with equivalence class $[K]$. We call the equivalence class $[K]$ a **knotted surface**.

Definition

We say that a surface **unknotted** if it is either the unknotted two-sphere U , or a finite connected sum of copies of the positive unknotted projective plane P^+ , the negative unknotted projective plane P^- , or the unknotted torus, T .

Connected Sums

But what is a connected sum?

Definition

Consider two tangle diagrams K_1 and K_2 . We can take a connected sum $K_1 \# K_2$ of the diagrams by detaching an endpoint on each diagram from the y -axis and connecting them. See the below figure.

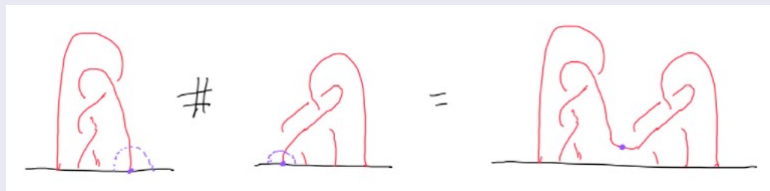


Figure: 8. Connected sum of two tangle diagrams.

Tri-plane Diagrams for Unknotted Surfaces

- Diagram for P^+



- Diagram for P^-



- Diagram for T



- Diagram for U



Bridge Number and Crossing Number

Definition

We define the **bridge number** $b(K)$ of a tri-plane diagram K as the number of strands in each picture. The bridge number of a knotted surface $b([K])$ is defined to be the minimal bridge number that appears for the given surface, i.e.

$$b([K]) = \min\{b(K') : K \sim K'\}.$$

Definition

We define the **crossing number** $c(K)$ of a tri-plane diagram K is the number of times that strands in all three diagrams cross. The crossing number of a knotted surface $c([K])$ is defined as the minimal crossing number that appears in some representation of the surface K , i.e. $c([K]) = \min\{c(K') : K' \sim K\}$.

Orientability

Definition

Given a tri-plane diagram K with b strands, to each endpoint we will associate either a $+$ or a $-$. We do so in such a way so that each strand has a $+$ at one endpoint and a $-$ at the other. We also ensure that the signs are consistent between all three diagrams. If all $2b$ signs in each diagram match and each strand receives one sign each, then we say that K is **orientable**, and we write $o(K) = 1$. Otherwise, $o(K) = 0$.

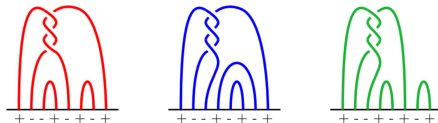


Figure: 7. The spun trefoil is orientable.

Writhe and Patch Numbers

Definition

Take a knot (or link) diagram and assign an orientation by moving along the knot with arrows. To each crossing where the arrow in the right direction is an over crossing, associate a $+1$. To each crossing where the arrow pointing right is an under crossing, associate a -1 . We define the **writhe** of K , $w(K)$ as the sum of these $+1$ and -1 which appear at each crossing.

Definition

We define the **patch numbers** p_{12} , p_{23} , and p_{31} as the number of components which appear in the unlinks for $K_1 \cup \overline{K_2}$, $K_2 \cup \overline{K_3}$, and $K_3 \cup \overline{K_1}$, respectively.

Two More Invariants

Definition

We define the **Euler characteristic** of a surface K as

$\chi(K) = p_{12} + p_{23} + p_{31} - b$, where p_{ij} are the patch numbers and b is the bridge number.

Definition

We define the **normal Euler number** of a surface K as

$e(K) = w(K_1 \cup \overline{K_2}) + w(K_2 \cup \overline{K_3}) + w(K_3 \cup \overline{K_1})$.

Orientability, Euler characteristic, and normal Euler number are all surface invariants. That is, if $K \sim K'$ then $o(K) = o(K')$, $\chi(K) = \chi(K')$, and $e(K) = e(K')$.

Classifying Unknots

Work of Group 02

- Find invariants of unknotted surfaces.
- Find the tri-plane crossing number for all unknotted surfaces.

Some Preliminary Results

- **Remark** If an unknotted P is orientable, then $[P] = [U]$ or $[P] = [T^n]$; otherwise $[P] = [P^+]^n \# [P^-]^m =: P^{n,m}$.
- **Proposition 1** $c([P]) = 0$ iff $[P]$ is orientable.
- **Proposition 2**
 $[T \# P^+] = [P^+ \# P^- \# P^+], [T \# P^-] = [P^- \# P^+ \# P^-]$
- **Corollary** $|n - m| \leq c(P^{n,m}) \leq \max\{n, m\}$.
- **Proposition 3** $c([P \# P']) \leq c([P]) + c([P'])$.

Main Results

Theorem

$$c([P^{n,m}]) = \begin{cases} 1 & \text{if } n = m \\ |n - m| & \text{if } n \neq m \end{cases}$$

Proof.

- **Step I.** Prove $c([P^{n,m}]) = c([P^{m,n}])$
- **Step II.** Prove $c([P^{n,n}]) = c([P^{n,n+1}]) = 1$
- **Step III.** Apply Prop. 3:

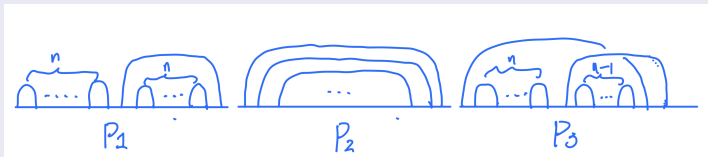
$$c([P^{n,n+k}]) \leq c([P^{n,n+1}]) + (k-1)c([P^{0,1}]) = 1 + (k-1) = k$$



Step II

Lemma

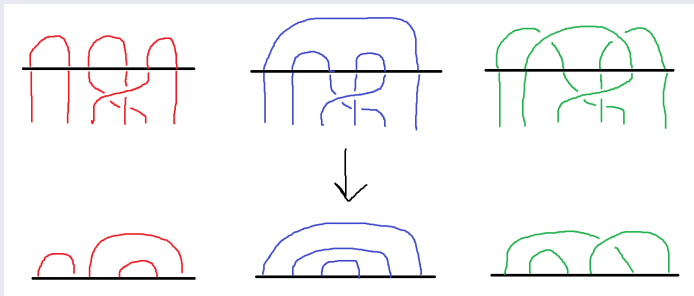
$\forall n \in \mathbb{N}_+, [P^{n,n}]$ has a diagram that looks like the following:



Step II

Proof.

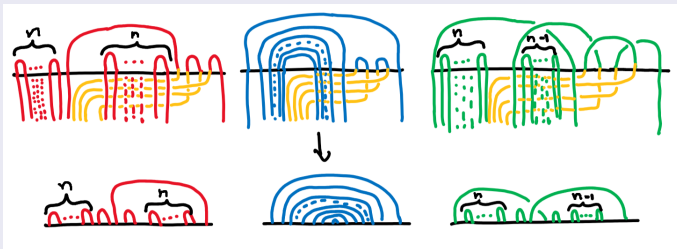
- Base case $n = 1$:



Step II

Proof.

- Inductive Step from n to $n + 1$:



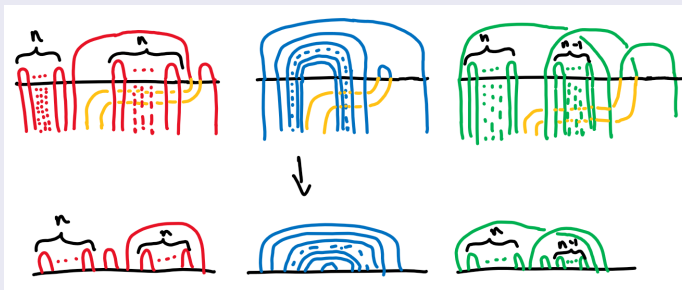
Step II

Corollary

$$c([P^{n,n+1}]) = 1$$

Proof.

- Perform braid moves as follows



Step III

Proof.

- $n > m$: $[P^{n,m}] = [P^{n,n+k}] = [P^{n,n+1}] \#^{k-1} [P^{0,1}]$
- By Proposition 3:

$$c([P^{n,n+k}]) \leq c([P^{n,n+1}]) + (k-1)c([P^{0,1}]) = 1 + (k-1) = k = |n-m|$$



Compiling a List of Known Knotted Surfaces

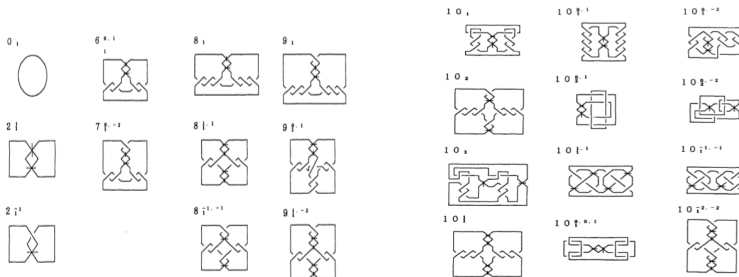
Compiling a Body of Knotted Surfaces

We want to compile a list tri-plane for knots of low complexity. By low complexity we mean low crossing number and low bridge number.

- Having a large corpus of examples is immensely helpful in the research process. They build intuition and are critical for testing conjectures.
- With knotted circles, it is easy to work with and come up with new knots.
- Unfortunately, this is not the case with knotted surfaces. Coming up with a list of distinct tri-plane diagrams for knotted surfaces is not trivial.

CH-Diagrams

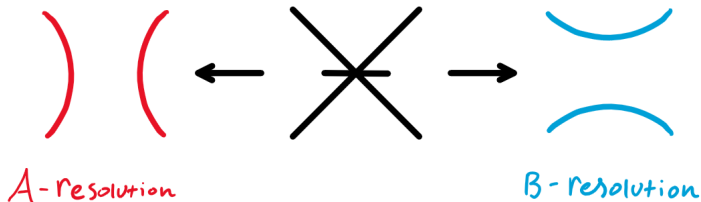
Luckily, there is already an enumeration of knotted surfaces with low complexity.



ch-diagrams for surface knots (Yoshikawa [2])

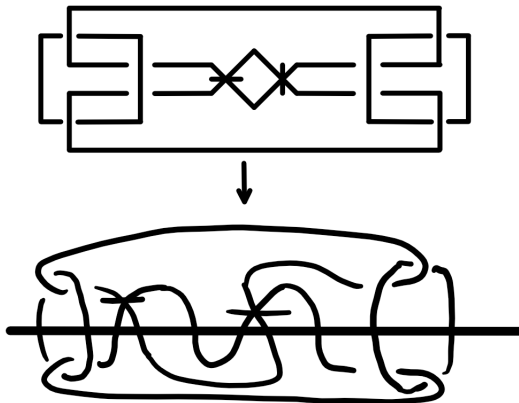
Converting CH-Diagrams

Let's convert a ch-diagram into a tri-plane diagram! The cross markings on the ch-diagrams represent a saddle point which is resolved as below.



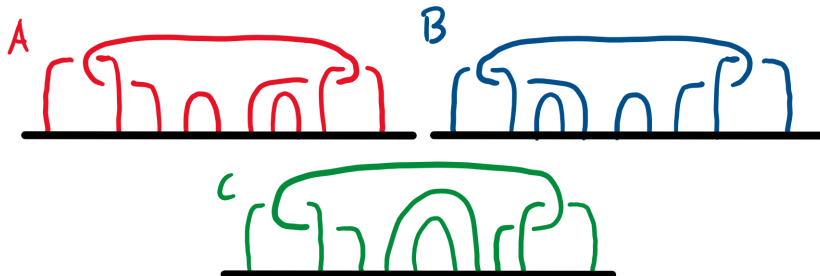
Converting CH-Diagrams

To resolve an entire ch-diagram, we first put it into bridge position:



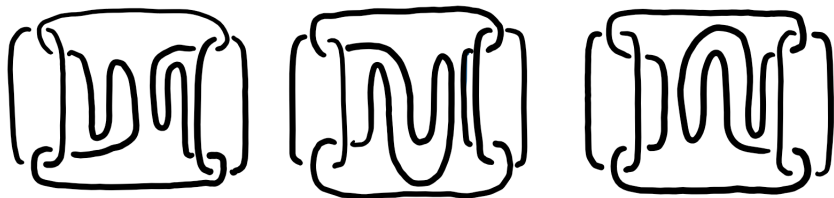
Converting CH-Diagrams

We then resolve the top of the ch-diagram into the tangles on top. The first one uses the A-resolution and the second uses the B-resolution. The bottom tangle is the bottom of the ch-diagram.



Converting CH-Diagrams

Finally, we must verify that $P_1 \cup \overline{P_2}$, $P_2 \cup \overline{P_3}$, and $P_3 \cup \overline{P_1}$ are diagrams for an unlink.



If we don't end up with unlink diagrams as we do here, we must go back to the beginning and put the ch-diagram into a better bridge position (usually one that results in a larger bridge number).

Conclusion

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References



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